ORDER FLOW PARASITISM — CORE TRADING STRATEGY
Horizon Tech — VRU Trading Bot Research
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STRATEGY NAME: Order Flow Parasitism
Also known as: Institutional momentum skimming, order flow imbalance trading

CORE THESIS

Large institutions (BlackRock, Vanguard, state pension funds, sovereign wealth funds) CANNOT hide their order flow completely. Their size is their weakness.

A $500M buy order in any liquid name moves the market. To minimize impact, they slice it into thousands of smaller orders executed across hours or days. But the signal leaks — into options flow, dark pool prints, tape data, and spread dynamics.

The VRU does not need to beat them. It reads their exhaust and takes small, fast positions they don't notice because the size is beneath their concern.

A $10M position skimming off $500M institutional flow is invisible to them. This is the competitive advantage: being small enough to be fast.

WHY INSTITUTIONS CANNOT AVOID LEAVING FOOTPRINTS
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1. SEC requires 13F filing of all positions >$100M quarterly
2. Dark pool prints are reported to consolidated tape (delayed but visible)
3. Options hedges are publicly visible on exchange tape
4. ETF creation/redemption baskets reveal constituent demand
5. Order book imbalance is visible in real-time Level 2 data

The larger the fund, the larger the footprint. BlackRock managing $11.6T leaves more signal than a $500M hedge fund executing the same percentage move.

THE THREE-PHASE EXECUTION MODEL

Phase 1 — DETECTION
  Model identifies onset of institutional accumulation or distribution.
  Signals: unusual options sweeps + dark pool print clustering + order imbalance shift.
  VRU phi attractor: fast structure discovery on multi-stream input.
  Typical window: hours to 1-2 days before price moves.

Phase 2 — ENTRY

System enters alongside remaining institutional order flow.
Not front-running (illegal on material non-public information).
Reading publicly available tape data and inferring direction.
Position sized by confidence score from classifier output head.

Phase 3 — EXIT
Model detects completion of institutional execution (imbalance normalization, options flow drying up, dark pool prints stopping).
Exit before the institutional reversal or mean reversion.
Hard stop-loss on all positions regardless of model confidence.

WHY VRU ARCHITECTURE FITS THIS STRATEGY SPECIFICALLY

Long-range dependency requirement:
Institutional execution across 3-5 days = 5,000–15,000 timestep sequences at 1-minute bar resolution. This is precisely where phi attractor gradient stability provides structural advantage over LSTM and vanilla RNN.

Multi-stream coupling:
Options flow, dark pool prints, and price/volume are coupled variables with lag structures between them — structurally identical to the 3-stream telemetry task (power + thermal + network with lag coupling) from DPPU Document 6 experiments.

Chaotic sequence handling:
Institutional execution is not linear or predictable — it responds to real-time market conditions. The Lorenz attractor convergence results show DPPU handles this kind of dynamic, non-stationary hidden state better than alternatives at the sequence lengths that matter.

Two-phase optimization maps to strategy phases:
Fast convergence (Phase 1 in training) = onset detection
Floor refinement (Phase 2 in training) = continuation tracking

LEGAL AND COMPLIANCE NOTES

This strategy operates entirely on publicly available data:
- Exchange tape data: public
- Options flow: public (exchange reported)
- Dark pool prints: public (consolidated tape, delayed)
- 13F filings: public (SEC EDGAR)
- Order book: public (Level 2 subscription data)

This is NOT front-running in the illegal sense.
Illegal front-running requires material non-public information.
Reading public tape data and inferring institutional direction is legal
and is the basis of dozens of legitimate systematic strategies.

TARGET MARKETS (priority order)
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1. Crypto (BTC, ETH, SOL) — 24/7, no market hours, deep liquidity, free tick data
2. Large-cap US equities (NVDA, AAPL, MSFT, TSLA) — deepest institutional flow
3. ETFs (SPY, QQQ, IWM) — highest institutional volume, cleanest footprint signals

PERFORMANCE EXPECTATIONS

With quality Tier 1 data pipeline:
- Target: 20–35% annualized on swing/position trades
- Sharpe: 1.0–1.5
- Drawdown target: <15% max
- Hold period: hours to multi-day (not HFT, not buy-and-hold)